

Hilbert Curve-Based Attention Enabling Topology-Preserving Image Tensor Representation for Semantic Segmentation Network

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Abstract

Drone-based building defect segmentation remains challenging due to complex surface textures and illumination variations. We propose *TPSegformer*, a topology-preserving segmentation framework that mitigates mis-segmentation in such scenarios. Its decoder incorporates a Hilbert curve-based topology-preserving mechanism to maintain spatial continuity and boundary precision during category layer computation. A lightweight multi-scale fusion module enhances semantic representation, while global context modeling strengthens holistic perception. Experiments on the building defect dataset show that *TPSegformer* outperforms existing segmentation methods, achieving 80.77% mIoU and 90.22% Acc. On the *Dac10k* dataset, it maintains strong generalization, reaching 44.27% mIoU and 60.32% Acc across diverse materials and defect types. The code is available at <https://github.com/mumu-k/TPSegformer>.

1. Introduction

Despite progress in semantic segmentation for autonomous driving, medical imaging, and Unmanned Aerial Vehicle (UAV) applications, adapting these methods to building defect inspection remains challenging. Building surfaces captured by UAVs often contain heterogeneous materials, complex structures, and drastic illumination or viewpoint changes, leading to mis-segmentation of intertwined defects such as cracks, spalling, and moss growth [16, 22]. For instance, stone textures can mimic cracks under certain lighting, causing false detections. Konior et al. [13] revealed strong material-defect correlations, yet most models overlook these relationships, limiting robustness and generaliza-

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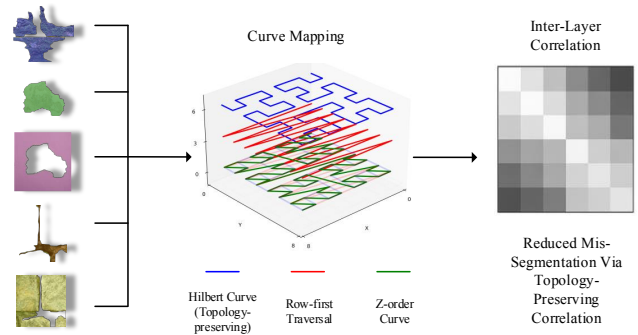


Figure 1. **Comparison of Spatial Traversal Strategies.** Three mapping methods—Hilbert Curve, Row-first, and Z-order—are compared. The Hilbert Curve best preserves image topology, reducing mis-segmentation.

tion in real-world UAV inspection scenarios.

To mitigate mis-segmentation, recent studies have enhanced semantic segmentation through architectural optimization. Some improve decoders via multi-stage up-sampling and encoder-decoder fusion for refined predictions [1, 15, 20], but their complexity and high computation limit UAV deployment. Others strengthen context modeling using advanced backbones [26] or feature refinement modules [35]. More recently, self-attention mechanisms have achieved strong contextual modeling performance [2, 12, 37], yet most rely on row-major traversal for feature unfolding, disrupting spatial continuity and weakening structural perception.

To address these challenges, we propose *TPSegformer*, a lightweight yet precise segmentation network for UAV-based building defect analysis. It integrates a Swin Transformer-Tiny backbone [17] with a custom lightweight decoder (TPDecoder) to balance accuracy and efficiency. TPDecoder fuses multi-scale features to enhance semantic consistency and employs a self-attention mechanism to

model inter-category dependencies, reducing class confusion. To preserve spatial structure, a Hilbert curve-based indexing strategy is introduced for dimensionality reduction, maintaining topological continuity during feature transformation, as illustrated in Fig. 1. Compared with traditional traversal methods such as the Z-order curve, this approach yields superior spatial coherence and robustness in complex defect segmentation scenarios.

Finally, given the current scarcity of datasets in the field of structural defect segmentation for buildings, this study reconstructs a segmentation dataset based on the BD3 architectural defect classification dataset[14]. The new dataset covers three typical defect types (cracks, peeling, and moss) as well as two major material categories (lime and stone). Using this dataset, we conducted systematic evaluations of TPSegformer, demonstrating its robustness, accuracy, and generalization capability in real-world architectural environments.

Our main contributions are as follows:

- **Topology-Preserving Lightweight Decoder (TPDecoder):** A novel decoder combining semantic enhancement and high-low feature fusion, integrated with a Hilbert curve-based traversal strategy that preserves spatial topology and strengthens inter-class relationships via attention modeling. Coupled with the Swin Transformer backbone, it enables high-accuracy, low-latency segmentation suitable for UAV-based building defect detection.
- **Comprehensive validation of Hilbert curve advantages:** Both theoretical and experimental analyses demonstrate that the Hilbert curve better maintains spatial continuity and enhances feature correlation compared to conventional row-major traversal, improving segmentation robustness and generalization.
- **Development of a high-quality building defect segmentation dataset:** Based on the BD3 dataset, a new segmentation set is constructed with labeled defect categories (cracks, spalling, moss) and material classes, providing a reliable benchmark for building defect analysis.

2. Related Work

2.1. Segmentation Decoder

Decoder design plays a key role in reconstructing fine-grained segmentation outputs from compressed encoder features, primarily addressing feature reconstruction, detail restoration, and multi-scale contextual fusion.

Feature reconstruction progressively restores high-resolution predictions from low-resolution features using transposed convolution or interpolation. For instance, TransUNet [2] integrates Transformer-based context modeling with transposed convolutions to enhance spatial resolution.

Detail restoration relies on skip connections to recover spatial information from shallow features. U-Net [20] first

adopted this idea, while Zhang et al. [33] introduced dual skip connections via a Dual-Skip Upsampling Fusion Module (DSUFM) to improve feature propagation.

Multi-scale contextual fusion integrates local and global cues. PSPNet [35] employs pyramid pooling, DeepLab [3–5] uses dilated convolutions [32], and HRNet [26] maintains multi-resolution representations through parallel branches. Recent Transformer-based methods such as ViT [7] and SegFormer [28] achieve similar effects via hierarchical token embeddings and MLP-based fusion.

Although effective, these decoders struggle with long-range dependency modeling, motivating the integration of self-attention mechanisms in recent works.

2.2. Self-Attention

Self-attention models global dependencies by computing spatial correlations[34]. Originally introduced in the Transformer [25], it has been widely adopted in computer vision to address the limitation of CNNs in capturing long-range dependencies.

However, its quadratic complexity with respect to input size incurs high computational cost on dense feature maps. To alleviate this, Tay et al. [23] proposed sparse attention to restrict computations to local or semantically relevant regions, while Lin et al. [15] adopted a shifted-window strategy for efficient cross-region interaction.

Later studies focused on improving efficiency and feature discrimination. Zhang et al. [38] developed the Attention Awareness Module (AAM) to emphasize informative regions, and Yin et al. [30] introduced Regional Self-Attention (RSA) with a dual-decoder structure for finer semantic distinction.

Nevertheless, few approaches preserve spatial topology during attention computation, which is crucial for accurate defect localization in complex structural environments.

2.3. Space-Filling Curves

Space-filling curves establish a mapping between one-dimensional sequences and multi-dimensional spaces. In 1890, Peano introduced the first such curve, followed by Hilbert, who proposed the more practical Hilbert curve [11]. Later, the Z-order curve [18] achieved spatial traversal by interleaving binary coordinates, becoming a widely adopted indexing strategy for high-dimensional data mapping.

Owing to their spatial continuity and topology-preserving properties, space-filling curves have been extensively applied in image processing. Tsinganios et al. [24] transformed sEMG signals into images via the Hilbert curve for CNN-based gesture recognition, while other studies [6, 21] employed similar mappings in image scrambling and encryption to enhance structural integrity and security.

In semantic segmentation, Zheng et al. [36] first incorporated the Hilbert curve into the self-attention mechanism,

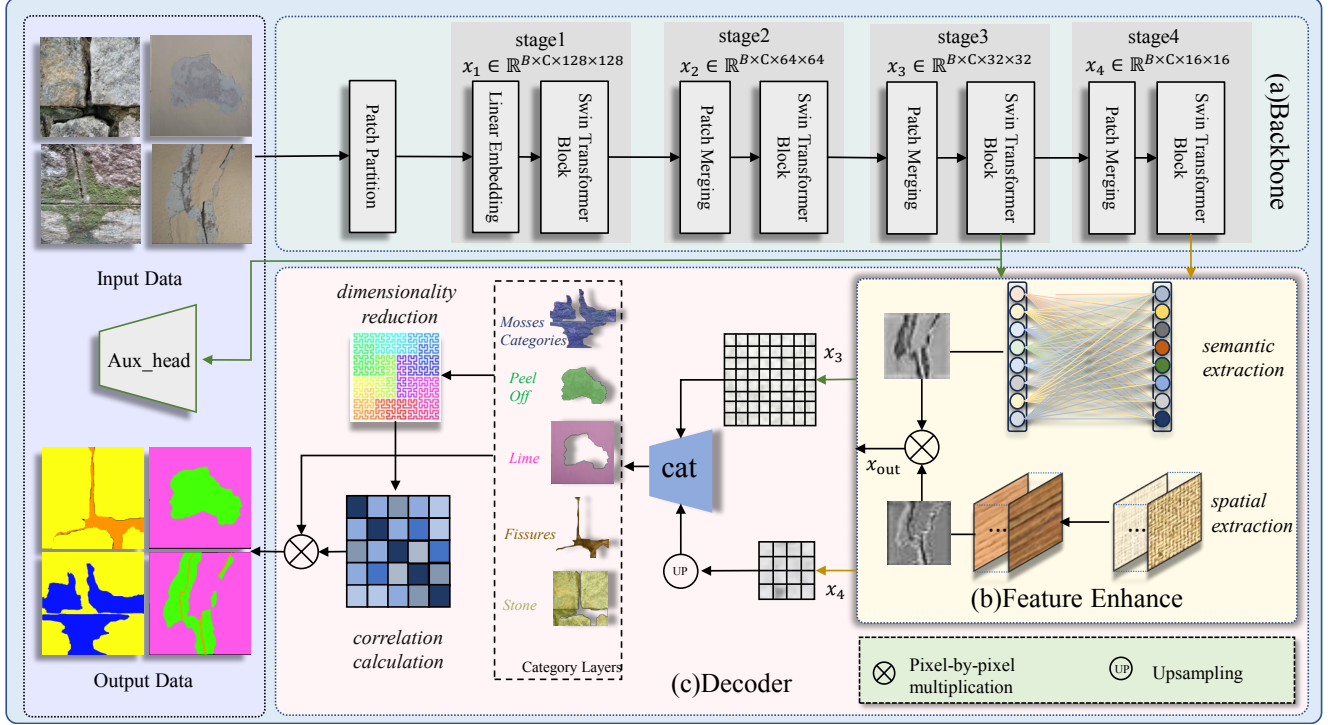


Figure 2. **Overview of the proposed segmentation network architecture.**(a) Swin-Tiny is employed as the backbone of the segmentation network for multi-scale feature extraction. (b) Feature enhancement: in the early stage of decoding, a dual-branch structure is used to separately enhance the semantic and spatial dimensions of the feature maps. (c) TP Decoder: multi-resolution feature maps are fused and then used to compute the correlations among category-specific layers.

replacing traditional row-major traversal to improve topology modeling. However, their work lacked a systematic comparison of different curve types and a theoretical analysis of their effects on attention, leaving room for further exploration.

In summary, existing work treats decoder design, attention modeling, and spatial encoding separately, leaving the joint issues of multi-scale fusion, inter-class correlation, and topology preservation unresolved. TPSeg-former addresses these gaps through semantic enhancement, topology-aware attention, and curve-based indexing for robust building defect segmentation.

3. Topology-Preserving Segmentation Network

3.1. Network Framework

The complete architecture of the segmentation network is illustrated in Fig. 2.

The proposed segmentation network is designed with three key principles—accuracy, lightweight architecture, and computational efficiency—to enable effective semantic segmentation of UAV-acquired imagery.

For feature extraction, the Swin Transformer-Tiny variant is adopted as the backbone to balance representational

accuracy and efficiency. As a hierarchical vision transformer, Swin partitions feature maps into non-overlapping and shifted windows, effectively reducing computational complexity while enhancing multi-scale representation.

Compared with conventional CNN-based backbones, it demonstrates stronger capability in modeling long-range dependencies and capturing global contextual information, thereby improving segmentation accuracy.

The backbone outputs four feature maps through successive downsampling: $X_1 \in \mathbb{R}^{B \times C \times 128 \times 128}$, $X_2 \in \mathbb{R}^{B \times C \times 64 \times 64}$, $X_3 \in \mathbb{R}^{B \times C \times 32 \times 32}$, $X_4 \in \mathbb{R}^{B \times C \times 16 \times 16}$, where B and C denote batch size and channel number, respectively.

To enhance decoding performance, a dual-branch feature enhancement module separately extracts semantic and spatial information, fusing them via pixel-wise multiplication to strengthen meaningful representations. In the decoder, a multi-resolution fusion strategy integrates enhanced features from X_3 and X_4 , recovering fine-grained edges and generating class-specific layers. Correlations among these layers are then computed to model inter-class relationships. Finally, a topology-preserving dimensionality reduction based on space-filling curves maintains structural consistency during correlation computation, reinforcing spatial

continuity and improving attention modeling quality.

3.2. Topology-Preserving Decoder

TPDecoder consists of two main stages: the feature enhancement stage and the decoding and prediction stage.

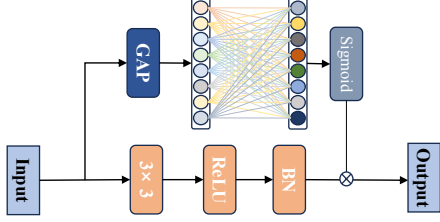


Figure 3. **Dual-Branch Feature Enhancement.** The dual-branch design jointly captures global semantic and local spatial cues, enhancing the expressiveness of intermediate features.

Feature Enhancement. After feature extraction by the backbone, fine-grained details may be lost, degrading segmentation accuracy. To mitigate this, we introduce a feature enhancement module based on ECANet [27], augmented with an additional spatial branch. The module jointly fuses channel attention and spatial context, enabling the network to focus on key regions, as illustrated in Fig. 3.

The channel attention branch follows ECANet’s design. Global average pooling (GAP) first generates a channel-wise descriptor, which is passed through a 1D convolution and Sigmoid activation to obtain channel weights:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W X_c(i, j), \quad c = 1, 2, \dots, C, \quad (1)$$

$$w = \sigma(\text{Conv1D}(z)), \quad (2)$$

where X denotes the input feature map and σ is the Sigmoid function.

The spatial branch applies a 3×3 convolution, followed by ReLU and batch normalization, to extract local texture and edge features often missed by channel attention:

$$F_s = \text{BN}(\text{ReLU}(\text{Conv3} \times 3(X))), \quad (3)$$

the final output is obtained via element-wise multiplication:

$$X_{out} = F_s \odot w, \quad (4)$$

where $\odot$ denotes channel-wise multiplication.

This fusion retains global semantic cues while reinforcing local structure representation, enhancing feature expressiveness and discriminability.

Decoding and Prediction. The proposed decoding framework comprises two modules: the High-Low Feature Fusion Module (HLFF) and the Lightweight Correlation Computation Module (LCCM), as shown in Fig. 4. Together, they fuse multi-resolution features and enhance semantic consistency for precise segmentation.

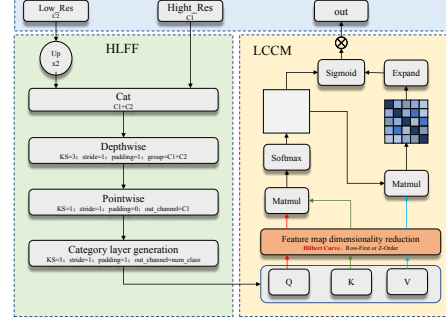


Figure 4. **Topology-Preserving Decoder.** The dual-module design fuses multi-scale features and models topology-aware correlations for precise and robust segmentation.

The HLFF module takes a high-resolution feature map $X_h \in \mathbb{R}^{B \times C_1 \times H_1 \times W_1}$ and a low-resolution feature map $X_l \in \mathbb{R}^{B \times C_2 \times H_2 \times W_2}$ from the backbone. The low-resolution feature is upsampled to the same size as X_h and concatenated along the channel dimension, forming $X_{cat} \in \mathbb{R}^{B \times (C_1+C_2) \times H_1 \times W_1}$. To remain lightweight, fusion is implemented using depthwise separable convolution—a 3×3 depthwise convolution for spatial extraction followed by a 1×1 pointwise convolution for channel compression. The fused features are then used to generate class-specific feature layers through a 3×3 convolution.

The LCCM introduces correlation modeling among these class-specific layers. The HLFF output is processed by three parallel convolutional branches to obtain query (Q), key (K), and value (V) maps, followed by dimensionality reduction. Three mapping strategies are explored: Hilbert curve traversal, row-first traversal, and Z-order traversal :

$$\mathcal{H}(x, y) = \sum_{k=0}^{n-1} 2^{2k} f_k(\text{bit}(x, k), \text{bit}(y, k)), \quad (5)$$

$$R(x, y) = x \times W + y, \quad (6)$$

$$\mathcal{Z}(x, y) = \sum_{k=0}^{n-1} [\text{bit}(x, k)2^{2k} + \text{bit}(y, k)2^{2k+1}], \quad (7)$$

all three map 2D coordinates $(x, y) \in [0, 2^n - 1]^2$ to a 1D index $h \in [0, 2^{2n} - 1]$. The Hilbert curve [11] preserves local spatial continuity, reducing topological distortion during flattening, while the other two mappings are used for ablation comparison.

After reduction, Q and K compute inter-category similarity, normalized by Softmax and weighted by V . The fused representation is refined by channel expansion and Sigmoid activation to produce attention weights, which modulate class-specific layers via element-wise multiplication, yielding the final enhanced segmentation output.

3.3. Auxiliary Supervision Mechanism

To enhance the discriminative capability of intermediate features, an auxiliary supervision mechanism is applied to the output feature map X_3 from the last stage of the Swin Transformer. This auxiliary branch adopts a lightweight FCN Head with five convolutional layers. Its output is up-sampled to the original image resolution via bilinear interpolation and optimized using a Joint Auxiliary Loss, which encourages semantic consistency and enhances boundary learning at the intermediate feature level.

The Joint Auxiliary Loss combines Dice Loss and an Edge-Aware Loss based on Sobel edge detection:

$$\mathcal{L}_{\text{aux}} = \lambda_{\text{dice}} \cdot \mathcal{L}_{\text{Dice}} + \lambda_{\text{edge}} \cdot \mathcal{L}_{\text{Edge}}, \quad (8)$$

where λ_{dice} and λ_{edge} are balance factors.

The Dice Loss mitigates class imbalance by maximizing the overlap between prediction and ground truth:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_{i=1}^N p_i g_i}{\sum_{i=1}^N p_i^2 + \sum_{i=1}^N g_i^2 + \epsilon}, \quad (9)$$

where p_i and g_i denote predicted probability and one-hot ground truth, and $\epsilon = 10^{-3}$ prevents division by zero.

To improve edge localization, the Edge-Aware Loss uses the Sobel operator to extract gradient magnitude maps:

$$\mathcal{L}_{\text{Edge}} = \alpha \cdot \|\nabla p - \nabla g\|_1, \quad (10)$$

$$\begin{aligned} \nabla p &= \sqrt{(p \otimes K_x)^2 + (p \otimes K_y)^2}, \\ \nabla g &= \sqrt{(g \otimes K_x)^2 + (g \otimes K_y)^2}, \end{aligned} \quad (11)$$

where K_x and K_y are Sobel kernels, and $\otimes$ denotes depth-wise convolution.

This joint supervision explicitly guides the backbone to learn fine-grained structural and boundary information, stabilizing training and improving overall segmentation performance.

4. Experiment

All experiments were conducted on a Linux workstation with an NVIDIA GeForce RTX 4090 GPU (24 GB), Python 3.8, and PyTorch 2.0.0 + CUDA 11.8. Model training used the AdamW optimizer with an initial learning rate of 6×10^{-5} , $\beta = (0.9, 0.999)$, and a weight decay of 0.01. A parameter-wise configuration was adopted for fine-grained optimization. The positional encoding and normalization layers were assigned zero weight decay to preserve stability, while decoder parameters were given a 10× learning-rate multiplier to accelerate convergence. This setup effectively stabilizes training and improves overall learning efficiency.

Table 1. Analysis of topological consistency via locality loss.

Method	Order					
	2	3	4	5	6	7
Row-First	42	406	3630	30814	254142	2064766
Hilbert Curve	15	63	255	1023	4095	16383
Z-Order	30	190	1086	5758	28926	139774

Table 2. Runtime comparison of 2D dimensionality reduction methods (seconds).

Method	Size				
	8×8	16×16	32×32	64×64	128×128
Row-First	0.000000	0.000000	0.000000	0.000998	0.009974
Hilbert Curve	0.000470	0.000999	0.005985	0.032912	0.130650
Z-Order	0.000000	0.000997	0.002991	0.010970	0.058365

4.1. Comparison of Spatial Traversal Strategies

This experiment compares three dimensionality reduction strategies—Row-First, Hilbert Curve, and Z-Order traversal—in terms of topological preservation and computational cost during the flattening of 2D images.

To evaluate spatial information loss, a locality loss metric is adopted, which measures the Euclidean distance between adjacent elements in the flattened sequence relative to their 2D positions:

$$\mathcal{L}_{\text{local}} = \sum_{i=1}^{N-1} \|p(i) - p(i+1)\|^2, \quad (12)$$

where $p(i)$ denotes the coordinate of the i -th pixel in the original 2D space.

As shown in Table 1, the Hilbert Curve achieves the lowest locality loss, indicating superior preservation of 2D spatial continuity. The Z-Order method performs moderately, while Row-First shows the weakest topological consistency.

As shown in Table 2, Row-First achieves the fastest runtime, while the Hilbert curve has higher computational cost. Overall, Hilbert traversal offers the best topological preservation with moderate overhead, making it suitable for low-resolution feature maps where structural integrity is critical and the time cost remains acceptable.

4.2. Comparison with Existing Methods

The segmentation dataset used in this study is reconstructed from the BD3 classification dataset, split into training, validation, and test sets in a 5:1:1 ratio. To evaluate segmentation performance, TPSegformer is compared against six representative networks—CCNet [12], SegFormer [28], BiSeNetV2 [31], DDRNet [19], PIDNet [29], and DSNet [9]. Quantitative results are shown in Table 3.

As shown in Table 3, TPSegformer consistently outperforms all baselines across categories, achieving the highest accuracy in fissures, peel off, mosses, and lime classes. It attains 80.77% mIoU and 90.22% ACC, surpassing the best

Table 3. Performance comparison of different segmentation networks on BD3. Abbreviations: Fiss. — Fissures, Peel — Peel Off, Moss. — Mosses.

Method	Fiss.	Peel	Moss.	Lime	Stone	mIoU	ACC
CCNet[12]	53.86	81.21	60.03	94.83	70.90	72.17	82.53
SegFormer[28]	50.60	72.49	57.35	95.99	93.72	74.03	86.43
BiSeNetV2[31]	45.11	65.43	58.70	94.43	78.00	68.33	77.50
DDRNet[19]	48.68	44.99	54.95	87.81	78.36	62.96	88.49
PIDNet[29]	55.90	78.60	55.37	95.07	79.71	72.93	84.27
DSNet[9]	48.17	64.73	32.81	93.22	78.73	63.53	80.91
Ours	59.98	91.11	72.58	96.71	83.44	80.77	90.22

Table 4. Comparison of Flattening Curves within TPSEgformer.

Method	mIoU (%)	ACC (%)
Row-First	75.58	86.37
Z-Order	77.89	88.00
Hilbert Curve	79.49	89.51

competitor by a clear margin. This improvement stems from the synergy between the High-Low Feature Fusion Module and Layer-wise Correlation Computation Module, which jointly enhance multi-scale feature integration and spatial topology preservation.

Although SegFormer achieves the best result on the stone category (93.72%), its overall performance is less stable. BiSeNetV2 and DDRNet perform poorly on complex textures (e.g., peel off), while DSNet exhibits weak generalization with only 63.53% mIoU, confirming the robustness advantage of TPSEgformer in complex multi-class segmentation scenarios.

4.3. Ablation Experiments

In this section, we analyze the effectiveness of the key components of TPSEgformer, focusing on: (1) the impact of different dimensionality reduction curves in TPDecoder; (2) the adaptability of TPDecoder with various backbone networks; and (3) the influence of hyperparameter settings in the Joint Auxiliary Loss.

Comparison of Spatial Flattening Methods within TPDecoder. We evaluate the segmentation performance of TPDecoder using three flattening strategies—Row-First, Z-Order, and Hilbert curve traversal—to analyze the impact of spatial topology preservation. The backbone is fixed as Swin Transformer-Tiny, and the cross-entropy loss is used.

As shown in Table 4, the Hilbert curve achieves the highest mIoU and ACC, confirming that preserving spatial topology during flattening enhances attention modeling and segmentation accuracy.

Effect of Backbone Architecture on TPDecoder Performance. To assess the adaptability of TPDecoder, we integrate it with three representative backbones—HRNet,

Table 5. Evaluation of TPDecoder Integrated with HRNet, ResNet, and Swin Transformer.

Backbone	Variant	mIoU (%)	ACC (%)
HRNet[26]	HRNet18	78.29	83.63
	HRNet48	79.72	85.73
ResNet[10]	ResNet18	73.44	84.28
	ResNet34	78.00	87.91
	ResNet50	79.00	88.09
	ResNet101	78.22	86.87
Swin Transformer [17]	Swin-T	79.49	89.51
	Swin-B	80.42	88.57
	Swin-L	81.37	90.02

ResNet, and Swin Transformer—and evaluate their segmentation performance on the BD3 dataset.

As shown in Table 5, the Swin Transformer series achieves the best overall performance, followed by HRNet, while ResNet performs the lowest. Notably, Swin-Tiny attains results comparable to larger HRNet and ResNet variants, demonstrating an excellent balance between efficiency and accuracy. These results confirm the Swin Transformer as an effective and scalable backbone choice for TPSEgformer.

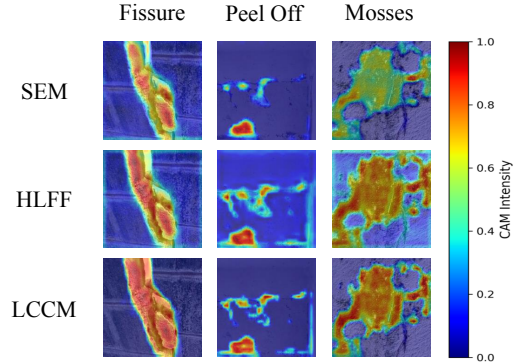


Figure 5. **Module-Level Focus on Specific Types of Building Defects.** Attention maps indicate that TPSEgformer increasingly concentrates on defect boundaries as decoding progresses from SEM to LCCM.

Experiments on Hyperparameter Settings in Joint Auxiliary Supervision. To determine the optimal configuration of the Joint Auxiliary Loss, we evaluated different combinations of λ_{dice} and λ_{edge} on the BD3 dataset using Swin Transformer-Tiny as the backbone. The results are shown in Table 6.

As shown in Table 6, introducing the Joint Auxiliary Loss consistently improves segmentation performance. The best results are achieved at $\lambda_{\text{dice}} = 0.5$ and $\lambda_{\text{edge}} = 1.0$, where TPSEgformer reaches 80.77% mIoU and 90.22% ACC, confirming the effectiveness of the proposed auxil-

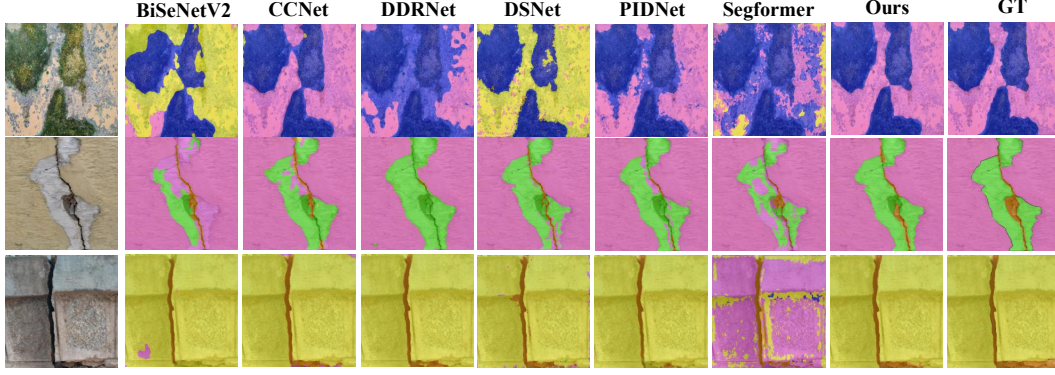


Figure 6. **Segmentation Results on the Test Set.** The visual comparison highlights the superior accuracy and generalization of TPSegformer, particularly in fine defect localization and edge preservation.

Table 6. Performance of Joint Auxiliary Loss under Different Hyperparameter Combinations.

λ_{dice}			λ_{edge}			Metrics (%)	
0	0.5	1	0	0.5	1	mIoU	ACC
✓			✓			79.49	89.51
	✓			✓		78.23	85.26
		✓	✓			79.83	85.48
✓				✓		79.82	89.4
	✓			✓		79.06	88.15
		✓		✓		79.22	87.52
✓					✓	79.84	86.46
	✓				✓	80.77	90.22
		✓			✓	79.56	87.1

iary supervision.

4.4. Visualization Experiments

We conduct three visualization experiments to demonstrate the effectiveness and practicality of TPSegformer. First, attention heatmaps are generated to visualize the focus distribution across TPDecoder modules. Second, segmentation outputs from multiple networks are visually compared to highlight TPSegformer’s advantages. Finally, real-world UAV video experiments are performed, where TPSegformer accurately segments building defects, validating its robustness and potential for practical deployment.

Heatmap Visualization. To illustrate how different TPDecoder modules attend to building defects, we visualize attention heatmaps for three defect types, as shown in Fig. 5.

As shown in Fig. 5, the decoder progressively refines its focus on defect regions. In the SEM stage, attention is coarse and some defects remain undetected. In the HLFF stage, attention intensifies and more regions are correctly identified. Finally, the LCCM stage exhibits highly concentrated activation around defect boundaries, yielding more precise localization and stronger category discrimination.

Segmentation Visualization Analysis. To further evaluate the visual performance of TPSegformer, we generate segmentation results on the test set of the BD3 dataset using the trained model weights.

As shown in Fig 6, most competing networks suffer from structural discontinuities and boundary ambiguities due to limited spatial topology preservation and inadequate multi-scale context integration. BiSeNetV2, DSNet, and SegFormer exhibit severe mis-segmentation, particularly in distinguishing lime and stone materials. DDRNet and PIDNet, despite enhanced edge supervision, fail to achieve precise boundary delineation, while CCNet performs poorly in crack localization.

TPSegformer addresses these limitations via a Hilbert curve-based topology-preserving mechanism that maintains spatial adjacency during feature flattening and a high-low feature fusion module that enhances multi-scale semantic representation. This design enables consistent global-local correlation modeling, yielding accurate defect localization, smooth edges, and improved robustness on complex building surfaces.

Field Test. To validate real-world applicability, a DJI Mavic 3 Pro drone was used to capture video footage of the experimental site. The recorded data were processed using TPSegformer to perform visual segmentation, as illustrated in Fig 7.

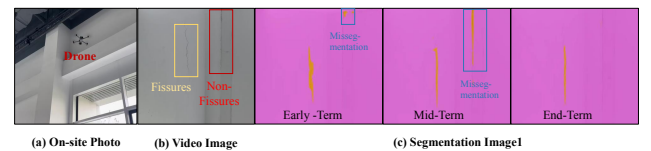
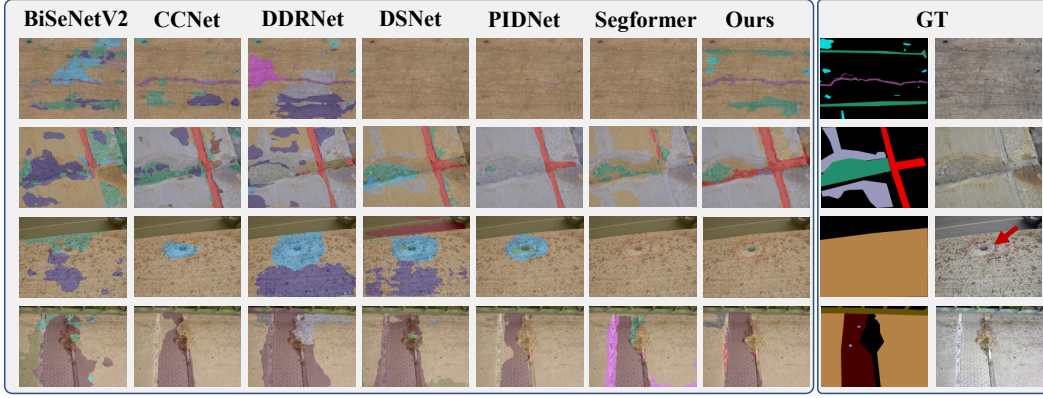


Figure 7. **On-site Validation using UAV-Collected Imagery.** Field experiments show that TPSegformer maintains stable segmentation performance under real UAV flight conditions, confirming its practical deployability.

Table 7. Performance Comparison of TPSegformer Against Six Networks on Dac10k Metrics.

Method	Crack	ACrack	Wetspot	Efflorescence	Rust	Rockpocket	Hollowareas	Cavity	Spalling	Graffiti	Weathering	Restformwork	ExposedRebars	Bearing	EJoint	Drainage	Equipment	JTape	WConcor	mIoU	ACC
DSNet[9]	0.00	8.64	9.82	20.03	32.51	2.04	37.01	23.48	25.22	53.31	48.57	17.67	3.87	59.74	36.97	25.48	66.15	21.8	7.41	26.30	38.32
PIDNet[29]	21.72	41.02	15.87	35.74	32.22	13.10	44.07	23.18	32.16	67.26	56.91	29.63	9.36	72.19	35.08	46.30	75.26	27.48	9.51	36.21	50.75
CCNet[12]	27.27	26.18	23.90	42.54	33.63	9.25	52.70	19.79	36.66	54.65	60.47	17.81	4.08	68.95	34.34	47.57	82.45	24.81	14.55	35.87	47.96
SegFormer[28]	10.35	30.87	19.96	38.26	27.61	0.04	56.29	27.02	29.08	55.69	55.50	3.38	0.00	72.03	49.80	43.38	77.95	13.70	1.39	31.96	42.28
BiSeNetV2[31]	8.04	24.10	13.02	27.65	28.57	0.71	39.68	22.17	29.93	42.55	51.26	9.14	0.10	49.77	18.19	21.45	71.90	15.34	13.26	25.62	36.60
DDRNet[19]	26.83	27.55	22.33	27.65	31.69	9.12	55.68	17.18	36.14	66.01	53.70	16.03	2.21	73.27	41.26	54.16	73.26	25.31	4.60	34.95	49.60
Ours	38.60	28.85	31.26	52.51	43.58	16.78	62.70	25.18	44.43	64.24	61.88	27.01	6.88	82.48	43.92	64.11	89.5	35.01	22.10	44.27	60.32

Figure 8. **Visualization of Dac10k Segmentation Results.** The results highlight the superior cross-scene generalization and semantic consistency of TPSegformer compared with existing segmentation networks.

As shown in Fig. 7, TPSegformer accurately detects cracks from UAV-captured footage. Although a dark painted region near the wall corner initially caused minor misclassification due to camera instability during focal adjustment, the model quickly recovered and precisely segmented the actual crack once the view stabilized. These results confirm the robustness and adaptability of TPSegformer in real-world UAV-based defect detection scenarios.

4.5. Generalization Experiment

To evaluate TPSegformer’s generalization capability, experiments were conducted on the Dac10k dataset [8], which contains 19 categories of structural defects (e.g., cracks, spalling) and common materials (e.g., concrete, asphalt, masonry) under diverse conditions. Table 7 compares the results with six benchmark networks.

TPSegformer achieves the highest performance in 12 of 19 categories, reaching 44.27% mIoU and 60.32% ACC, surpassing the second-best model by 8.06% and 9.57%, respectively. These results demonstrate its strong generalization and stability across diverse defect types and material.

As shown in Fig. 8, TPSegformer shows clear advantages in segmenting diverse structural defects. In the first row, it precisely detects fine cracks consistent with the ground truth (GT), whereas DSNet, PIDNet, and SegFormer fail to do so. In the second row, despite the JTape region being unannotated in GT, TPSegformer correctly perceives its structure, indicating strong structural awareness. In the third row, the concave area (red circle) is un-

labeled in GT but correctly identified as Spalling by TPSegformer, reflecting superior generalization and semantic understanding. In the fourth row, both TPSegformer and SegFormer segment the EJoint region accurately, but TPSegformer yields smoother and cleaner boundaries.

Overall, TPSegformer excels in fine-grained defect localization, boundary precision, and robustness in complex or unannotated regions, showing strong cross-domain adaptability in multi-material environments.

5. Conclusion

This paper investigates the relationship between architectural defects and construction materials and presents TPSegformer, a lightweight semantic segmentation network designed to achieve high segmentation accuracy with efficient computation. TPSegformer employs the Swin Transformer as its backbone for multi-scale feature extraction and introduces a lightweight TPDecoder that enhances and fuses features to generate class-specific maps. A Hilbert curve-based dimensionality reduction strategy preserves spatial topology while improving computational efficiency, and a self-attention mechanism models inter-class dependencies to mitigate missegmentation.

Field experiments validate the accuracy, robustness, and adaptability of TPSegformer in complex architectural environments. Future work will focus on further optimizing the model and deploying it on UAV platforms to support intelligent building inspection and defect analysis.

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